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PALAEOMAGNETIC EVIDENCE FOR LARGE-SCALE DEXTRAL MOVEMENT ALONG THE TROLLFJORD–KOMAGELV FAULT, FINNMARK, NORTH NORWAY

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Palaeomagnetic data from Late Precambrian dykes from the northern part of Varanger peninsula, north Norway, suggest a two-axis magnetization structure. The dominant component is considered to be syn- to late-tectonic and probably acquired at around 640 m.y. B.P. Superposed on this magnetization is a minor component which is compatible with the relative Lower–Middle Palaeozoic field; i.e. it was most likely imposed during the climax of the Caledonian orogenic movements in north Norway. The estimated relative Late Precambrian palaeopole cannot easily be reconciled with the European Late Precambrian polar path. This disagreement can be accounted for by assuming a post-magnetization dextral megashear, of the order of 500–1000 km, along the Trollfjord–Komagelv fracture zone. This type of displacement is in line with geological evidence and the palaeomagnetic reconstruction supports the long-held view of there having been continuity between the depositional environments of the Varanger Peninsula Barents Sea Group, the Eleonore Bay Group of east Greenland and the Hecla Hoek Formation of east Spitsbergen. The character and age of the horizontal displacement, post-640 to pre-500 m.y. B.P., is seen in conjunction with the opening up of the Iapetus Ocean and reactivation of ancient deep-seated fractures during both the spreading and the contraction phases of ocean development.

1. Introduction

Geological investigations on the Varanger peninsula, north Norway, over the past few years have added greatly to our understanding of this key area, situated at the junction between the Scandinavian–Caledonian and the Timanian–Baikalian orogenic belts. Reconnaissance studies and subsequent detailed mapping allowed Siedlecka and Siedlecki (1967, 1971, 1972) to establish a comprehensive stratigraphy, building partly on the earlier work of Holtedahl (1918), Føyn (1937) and Reading (1965), and permitted comparisons and suggested correlations to be made between this region and neighbouring areas in NW Russia (Siedlecka, 1975).

An important element in the geological development of Varanger peninsula is that of a major NW–SE-trending tectonic dislocation: the Trollfjord–Komagelv Fault Zone. This fracture divides the peninsula into two separate geological regions (Siedlecka and Siedlecki, 1967). Rocks southwest of the fault zone, in the Tanafjord–Varangerfjord region, range from Upper Riphean to Lower Cambrian in age with younger strata, up to Tremadocian, represented by the Digermul Group (cf. Fig. 1). This autochthonous succession, which contains two tillite formations, rests unconformably on a high-grade crystalline basement. For further details the reader is referred to Reading (1965), Siedlecka and Siedlecki (1967,

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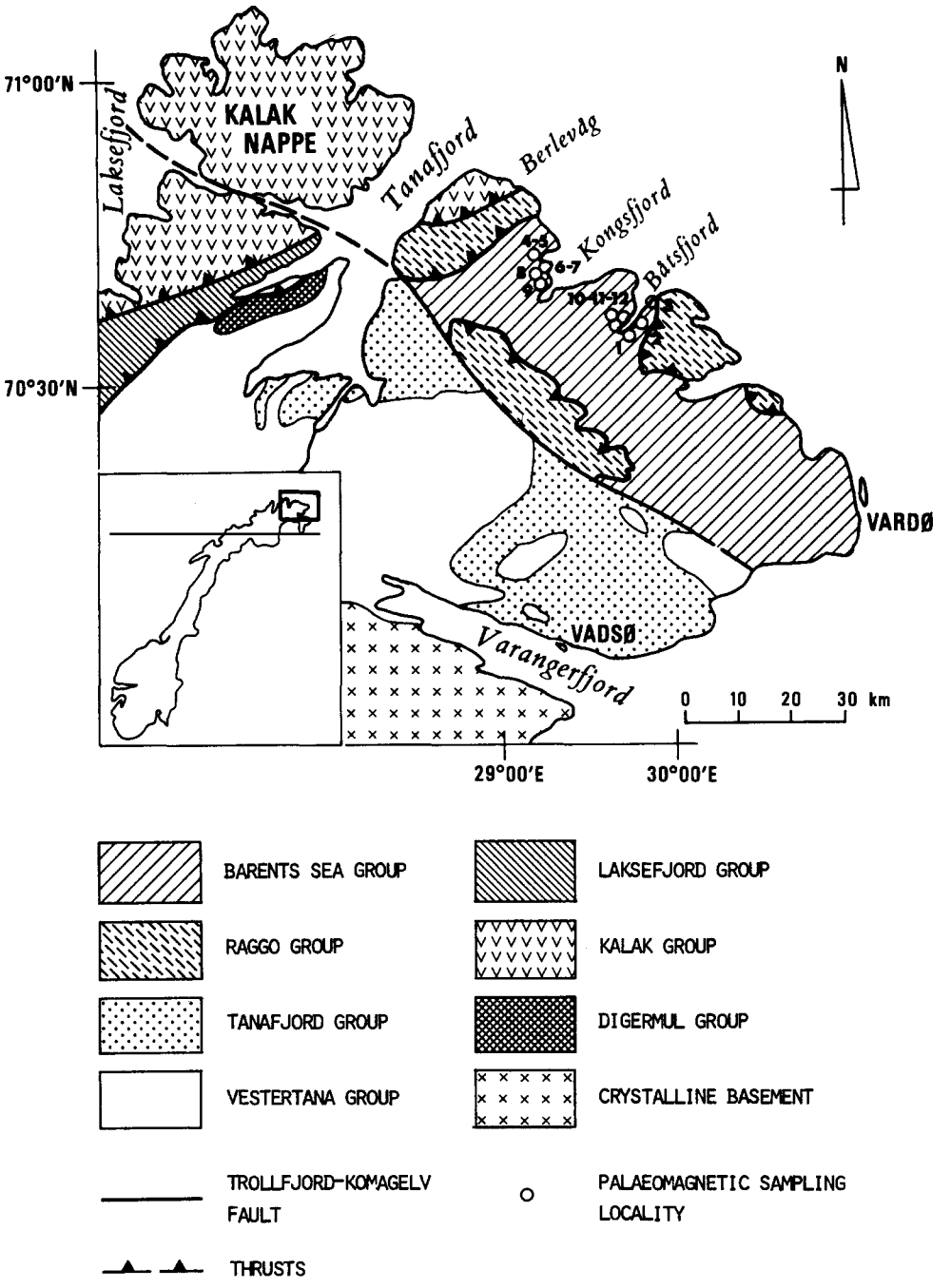


Fig. 1. Geological sketch map of the Varanger peninsula and adjacent areas.

1971, 1972), Banks et al. (1971) and Siedlecka (1975).

Two principal stratigraphic units have been described from northeast of the Trollfjord–Komagelv Fault Zone (Barents Sea region). These are the Barents Sea Group and the Raggo Group (Siedlecka and Siedlecki, 1967). The 9 km thick Barents Sea Group is considered to be of Riphean age and has

been divided into four formations: Kongsfjord (oldest), Båsnæring, Båtsfjord and Tyvjofjell Formations. The age and affinities of the Raggo Group have been the subject of some controversy with interpretations ranging from Riphean, and in part older than the Kongsfjord Formation (Siedlecka and Siedlecki, 1972; Siedlecka, 1973), to Vendian (Laird, 1972; Gayer and Roberts, 1973; Sturt et al., 1975). The most recent investigations in the type area have revealed that the two formations composing the Raggo Group are separated by a tectonic discontinuity, such that the name "Raggo Group" must now be rejected (Levell and Roberts, 1977).

Dolerite dykes intrude the various successions, particularly in the Barents Sea region where they locally attain swarm proportions. The most prominent dyke set in the Barents Sea Group is broadly coeval with the major fold-producing deformation and lowest greenschist facies metamorphic event; the dykes are parallel or subparallel to the fold axial surfaces, sometimes cutting obliquely across folds from limb to limb, yet carrying the axial planar cleavage which pervades the host metasediments. These dykes have been described as metadolerites (Roberts, 1975) to distinguish them from younger, uncleaved, post-folding dolerites. The metadolerites occur exclusively northeast of the Trollfjord–Komagelv Fault, and mostly in the Kongsfjord and Båsnæring Formations, while sporadically occurring uncleaved dolerite dykes are found on both sides of the dislocation. The strongly altered metadolerites are hereafter referred to as the "Kongsfjord dykes".

K/Ar age determinations for the younger dolerite dyke set denote an intrusion age of around 355 m.y. B.P. (Beckinsale et al., 1975). Data from the metadolerites, on the other hand, are inconclusive and unreliable giving ages ranging from 935–1946 m.y. B.P. (Beckinsale et al., 1975), probably due to the presence of excess Ar. These same authors have also sampled from supposedly uncleaved dykes cutting the Båtsfjord Formation (hereafter called the "Båtsfjord dykes"). The latter dyke set shows identical trend and structural relationships and a comparable chemistry to those of the metadolerites, and some of the dykes do in fact carry a weak cleavage which is axial planar to regionally developed folds (D. Roberts, unpublished data, 1976). The Båtsfjord dykes have given ages of around 640 m.y. B.P. These radiometric dates of Beckinsale et al. are in line with earlier suggestions that a major transcurrent movement must have occurred along the Trollfjord–Komagelv Fault Zone (Harland and Gayer, 1972; Roberts, 1972), in that the rocks intruded by the 640 m.y. B.P. (and possibly older) dykes occur exclusively north of this dislocation.

In the present paper we report some palaeomagnetic results from the Varanger peninsula Kongsfjord metadolerites and the Båtsfjord dolerites. The aim of the study was to contribute towards an understanding of the nature of the Trollfjord–Komagelv dislocation. Under favourable circumstances palaeomagnetism may allow calculation of the approximate magnitude of lateral movement, if present, as well as helping to resolve other problems such as the actual relevance of the suggested stratigraphic correlations across the fault zone.

2. Late Precambrian palaeomagnetism from the Varanger peninsula

The palaeomagnetic results to be reported here come from dykes in the Båtsfjord and Kongsfjord areas (Kjøde, 1976). Six dykes (thicknesses ranging between 3 and 5 m) have been studied in each area. The distribution of sampling localities is shown in Fig. 1. Though the dykes have clear baked contacts with their host rocks they show variably developed alteration features due to metamorphic processes, hydrothermal activity, fracturing and cleavage development, but the Båtsfjord dykes are generally rather fresh-looking as compared with those from the Kongsfjord area. The rocks proved to be too hard for field drilling, however, so that hand sampling was used throughout. For orientation of the 80 blocks collected both magnetic and sun compasses were used. As a detailed account of the palaeomagnetic and rock magnetic properties will be dealt with in a separate paper only the most important results are summarized here.

2.1. The Båtsfjord dykes

More than 75% of the opaque mineralogy is represented by titanomagnetite and its alteration products.

The rest is free ilmenite and sulphide (pyrite). The average "titanomagnetite" (TM) grain size is ~100 μm but grain sizes up to 1000 μm have been observed. Broad exsolution lamellae in some grains indicate a slow cooling and the further alteration of some of these lamellae into pseudobrookite suggests that at least some of the grains can be placed in the high-temperature oxidation classes IV and V (Ade-Hall et al., 1968). However, the predominant feature of these oxide grains is an extreme degree of fine-scale disintegration. In polished sections these bear close resemblance to granulation (cf. Ade-Hall et al., 1968). Many grains are more strongly magnetic along the edges (as revealed by the Bitter technique) than in the interior which may indicate processes of migration of ferrous Fe towards the rims, forming cation-deficient spinel phases on subsequent oxidation. In a number of specimens the TM groundmass has suffered a complete chemical disintegration while the ilmenite lamellae apparently still remain unaltered. According to Ade-Hall et al. (1971) this signifies that the temperature at which the low-temperature oxidation took place did not exceed about 250°C.

The NRM directions are relatively well-grouped with an overall mean direction of around: $D_m = 90°$, $I_m = +70°$. On alternating field (AF) demagnetization a major part of the remanence intensity has in general been randomized below about 300 Oe. A few specimens show a much higher stability upon AF treatment (at least 20–30% of the remanence intensity may remain after demagnetization in 1000 Oe). For the magnetically "hardest" specimens, the maximum demagnetizing field has been about 1500 Oe. On thermal demagnetization 73 out of 92 analysed specimens exhibit a pronounced blocking temperature range between 450 and 550°C, while the remaining ones define a more linear intensity decay pattern.

After removal of a low-stability component the specimens for which an underlying, more stable, remanence direction could be determined group very well in the same region as the majority of NRM directions (in this respect there is no difference between magnetically soft and magnetically hard specimens). In this group of specimens about 30% remain essentially stationary throughout the range of demagnetization. Both on thermal and AF treatment the last 15% or so of the remanence intensity is in general associated with directional swings either towards a northeasterly or towards a southwesterly direction. Since these systematic directional changes take place just before the onset of the erratic stage of demagnetization (where reliable results cannot be obtained) the determination of the associated palaeomagnetic directions can only be loosely indicated. However, based on some four specimens for which the latter directions (low-stability remanence removed) play a more dominant role than in the rest of the collection it appears that the weak remanence components define a nearly horizontal magnetization axis striking NE–SW.

If one disregards the directional changes that frequently take place on demagnetization at comparatively low intensities one ends up with distributions of remanence directions as shown in Fig. 2. It can be shown that both on specimen as well as on sample level the directions of magnetization conform to a Fisherian distribution (Fisher, 1953). Though the groupings are better defined in the present case than what one normally encounters in palaeomagnetic studies, there is a slight elongation of the distributions lining up with the trends of directional changes on demagnetizations (at low intensities). From the demagnetization studies the southwestward movement is twice as frequent (35 specimens) as the northeastward movement (16 specimens). This may indicate that on the whole the southwesterly component of the shallow magnetization axis is more important than the northeasterly one. Therefore, if any adjustment of the mean direction for the dominant magnetization of this formation, $D_m = 86°$, $I_m = +60°$, should apply, it is likely to be that of a slight northward shift.

Table I shows various palaeomagnetic parameters for different levels of observation. By analysis of dispersion (Watson and Irving, 1957) it can be shown that there is no significant between-dyke scatter. This may mean either that; (a) the Båtsfjord dykes were intruded over a short interval of time and therefore collected only a minor fraction of geomagnetic secular variation, or that (b) low-temperature alteration may have more or less completely erased the original TRM/TCRM magnetization and subsequently accumulated geomagnetic secular variation (through chemical processes) over a major time span. In the latter case the palaeomagnetic mean direction of each dyke may be expected to conform to that of an axial dipole

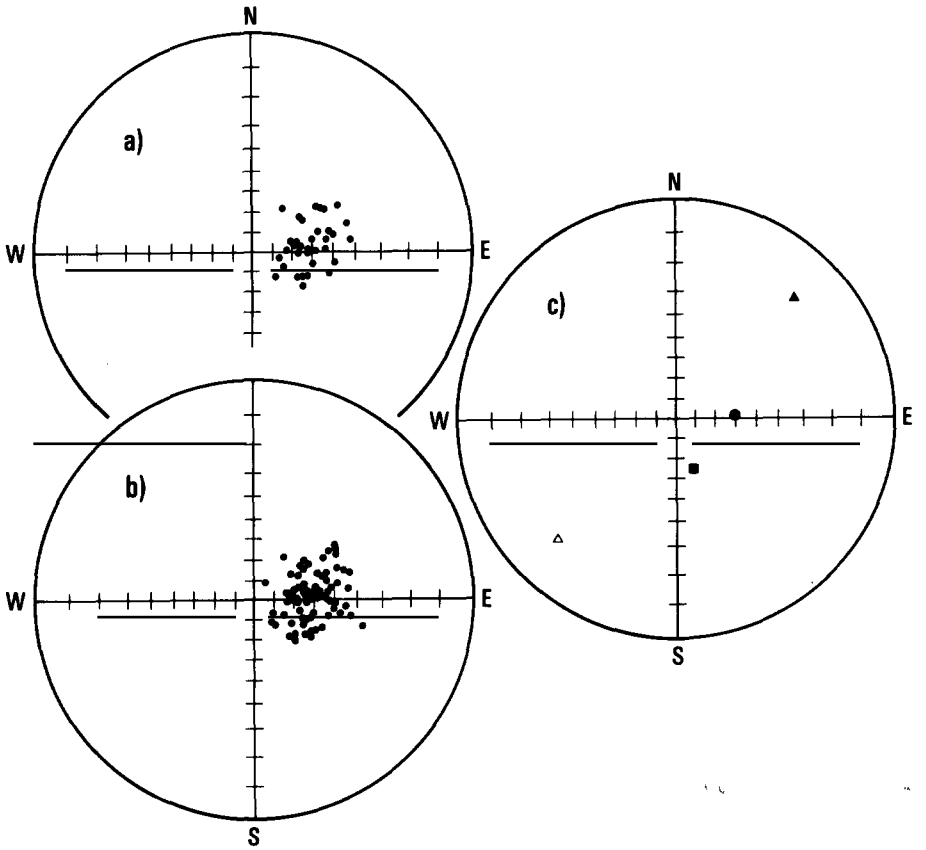


Fig. 2. Distribution of bulk directions of magnetization for the Båtsfjord dykes: (a) samples and (b) specimens. In (c) the Båtsfjord mean direction of magnetization (*circle*) is compared with the relative Late Precambrian field based on the overall polar mean from south Norway (*square*) (see Fig. 3 for references) and with the dominant Lower–Middle Palaeozoic field (*triangles*). Solid (open) symbols are on the lower (upper) hemisphere.

field and consequently the between-dyke scatter would be minimal. In the present case, where the intrusion seems to be syn-tectonic (see below), followed by extensive low-temperature ($T < 250°C$) alteration of the primary oxides, the second explanation seems to be by far the most likely alternative.

TABLE I

Palaeomagnetic parameters for the Båtsfjord dykes

Level of observation	N	R	k	α_{95} (°)	D_m (°)	I_m (°)	Pole	
							lat. (°N)	long. (°E)
Specimen	74	71.4	27.7	3.2	86.4	+59.9	39.1	107.0
Sample	34	33.0	32.3	4.4	84.5	+60.6	40.6	108.2
Dyke	6	5.95	92.9	7.0	87.0	+61.7	40.9	105.3

N = number of unit vectors; R, k and α_{95} = parameters according to Fisher (1953); D_m = mean declination; I_m = mean inclination.

2.2 The Kongsfjord dykes

The magnetomineralogy of these rocks is completely different from that of the Båtsfjord dykes. Titanomagnetite seems to have been present originally but these grains have suffered a nearly complete alteration, though some relics of altered ilmenite can still be observed. The only observed magnetic mineral is pyrrhotite. The grain sizes of this mineral vary from less than 5 μm to more than 1500 μm. J_s–T measurements show predominant inflexion temperatures between 325 and 400°C which is typical for pyrrhotite (Haraldsen, 1937). The bulk remanence intensity is considerably lower than for the Båtsfjord dykes, being mostly in the order of 10^{-5}–10^{-6} e.m.u./cm^3. The NRM directions are also much more scattered than those of the Båtsfjord dykes but the most dense grouping (constituting about 60% of all the results),

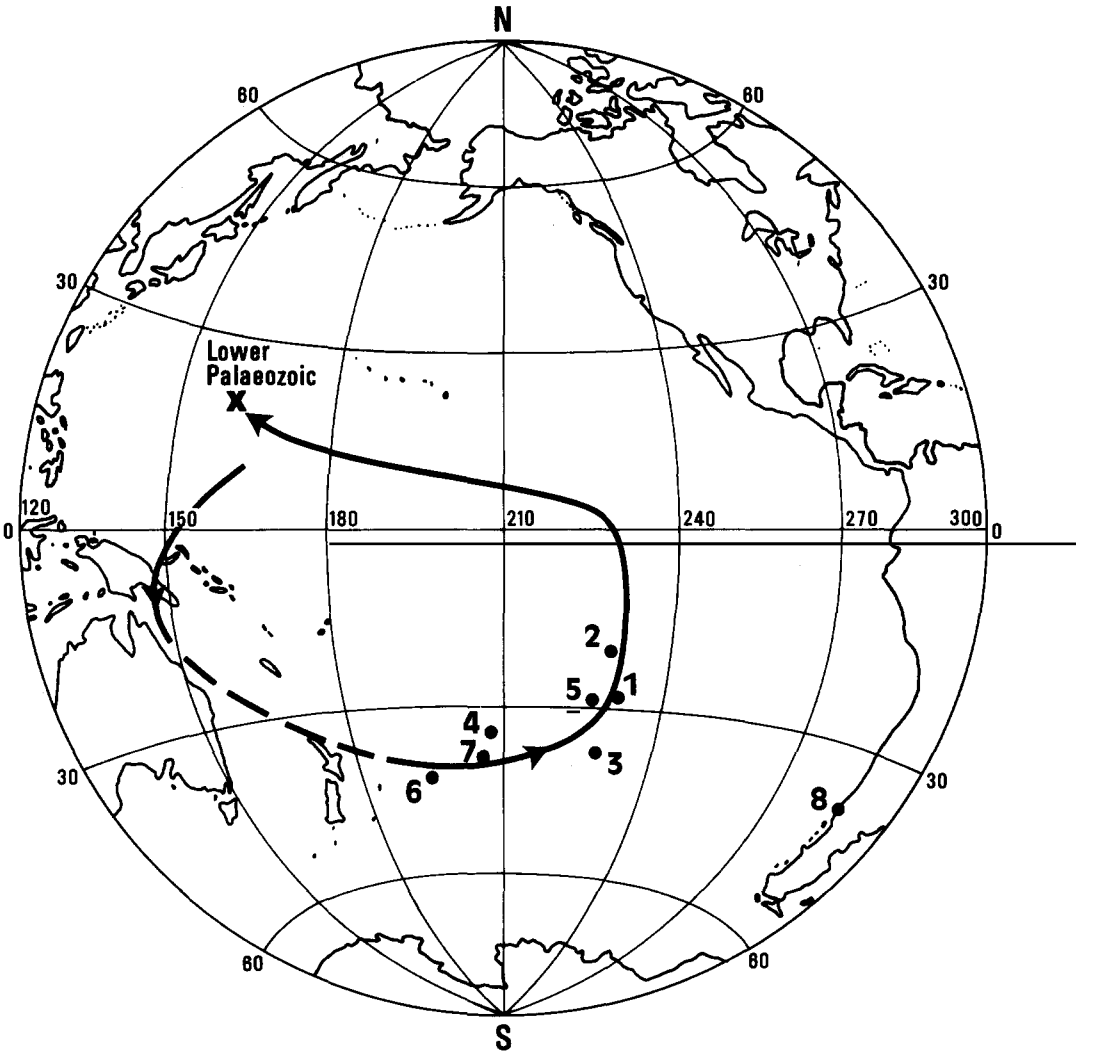


Fig. 3. The Båtsfjord pole (*8*) seen in conjunction with other Late Precambrian poles for NW Europe. These are: *1* = Storetvedt and Gidskehaug (1968); *2–4* = Poorter (1972); *5* = Stewart and Irving (1974); *6* = Murthy and Deutsch (1976); *7* = Murthy and Deutsch (1974). The ages of these poles are thought to range between ~950 and ~800 m.y. B.P. and are therefore probably slightly older than the Båtsfjord pole. The indicated polar path for the latest Precambrian is according to Spall (1973).

as well as the overall mean direction, is equivalent to the main cleaned palaeomagnetic direction of the latter sequence. On both thermal and AF demagnetization, however, stable palaeomagnetic directions were in general impossible to define. After a certain stage of systematic vectorial movement, but in most cases without establishing a terminal direction, the specimens came into a new stage where their intensity as well as their direction of magnetization varied erratically. Similar problems have recently been encountered in pyrrhotite-bearing rocks from the Oslo area (Pedersen, 1975). In fact, only a few specimens from the Kongsfjord dyke suite defined stable end-points on progressive demagnetization, although these do agree with the results as deduced from the Båtsfjord dykes.

Though there seems to be close resemblance in the palaeomagnetic structure of the two dyke sequences, only the Båtsfjord data qualify as a basis for the tectonic consideration that follows. The shallow magnetization axis that is inferred from the Båtsfjord dykes corresponds to the major Lower–Middle Palaeozoic geomagnetic axis relative to central and north Europe. This field axis seems to have existed at least from the base of the Cambrian (Storetvedt, 1973) and until it was discontinued at about Lower–Middle Carboniferous time (Storetvedt, 1968). In all probability this field was impressed in the rocks during the climax of the Caledonian orogeny in the region (530–500 m.y. B.P.). The main palaeomagnetic component, however, (obtained after partial demagnetization but disregarding the directional changes at low remanent magnetizations) does not fit with any known Phanerozoic palaeofield for Europe. Most likely this magnetization dates from latest Precambrian time.

The indicated similarity in the palaeomagnetic build-up of the Båtsfjord and Kongsfjord dykes supports recent geological and geochemical evidence (D. Roberts, unpublished data, 1976) that, despite the much higher degree of metamorphic and hydrothermal alteration in the Kongsfjord dykes, the two dyke suites may be considered as deriving from one and the same intrusive event. The palaeomagnetic similarity also suggests that the magnetization processes are syn- to late-tectonic, probably taking place at around 640 m.y. B.P. On the other hand, the palaeopole based on this suggested Late Precambrian magnetization (cf. Fig. 3) is not in agreement with the relevant polar path for Europe. We suggest that this has a geotectonic explanation which will be further elaborated upon below.

3. The palaeomagnetic data in a geological framework

The demonstration that the estimated palaeopole for the Båtsfjord dykes cannot easily be reconciled with the apparent polar-wander curve for the Late Precambrian of NW Europe leads one to consider a possible geotectonic explanation for the deviating direction of magnetization. This at once throws focus on the Trollfjord–Komagelv Fault as the likely structure along which a rotational movement may have been accomplished. Recognition of this dislocation as signifying a major tectonic element separating disparate sequences, the Tanafjord–Varangerfjord and Barents Sea regions, was first made by Siedlecka and Siedlecki (1967) in their excellent reconnaissance study of the Varanger peninsula. In this and later work (Siedlecka and Siedlecki, 1971, 1972; Siedlecka, 1975) the structure was described as fundamentally a thrust-fault with SW-directed movement. The complex history of the fault was emphasized, however, as was its probable southeastward extension across the Ribachiy–Sredniy peninsula in the U.S.S.R., and further into the Timans. On tectonic considerations Roberts (1972) concluded, on the other hand, that the major primary movement was of dextral transcurrent character, with only comparatively subordinate, and later, dis-slip components.

Speculating on the Late Precambrian–Early Palaeozoic cratonic and oceanic configurations in Arctic regions Harland and Gayer (1972) suggested that the Trollfjord–Komagelv Fault may represent an important structural line and possible plate boundary, along which two different sedimentary sequences had been juxtaposed by Silurian dextral transcurrent movement. The plates in question are the Baltic and Barents cratons. This reconstruction was later followed by Gayer and Roberts (1973) although these authors preferred to place the actual plate boundary to the north of the Varanger peninsula, regarding the Trollfjord–Komagelv Fault as a parallel major structure.

Recent K/Ar ages for dykes from the Varanger peninsula (Beckinsale et al., 1975) have tended to

confirm the view of the Trollfjord–Komagelv Fault as a major transcurrent fracture. Based on this radiometric data, the penetrative thermal and deformation event affecting Barents Sea Group rocks on the northern side of the fault zone was considered to be Late Precambrian in age, a good deal older in fact than the folding and cleavage which pervades the autochthon, post-Tremadocian, on the southern side of the fault zone, despite the similarities of fold style, metamorphic grade and cleavage development (cf. Roberts, 1972). As Beckinsale et al. (1975) conclude, this signifies that the crustal blocks on either side of the Trollfjord–Komagelv Fault were not in juxtaposition at 640 m.y. B.P. An upper limit for at least the bulk of the transcurrent movement along the fracture zone is Late Devonian since dykes of this age occur on both sides of the fault (Beckinsale et al., 1975; Roberts, 1975).

In the present context it is important at this stage to consider the Varanger situation in a wider perspective and to reiterate the long-held view of there having been a large measure of continuity between the depositional environments represented by the Riphean–Vendian sequences of east Greenland and eastern Spitsbergen (Kulling, 1934; Harland, 1969, 1973). Comparisons between southern and northeastern Spitbergen have also been documented (Birkenmajer, 1975). The succession in western Spitsbergen, however, appears to be quite different. Lithostratigraphic comparisons have also been made between the East Greenland Eleonore Bay Group sequence and the Barents Sea assemblage of the Varanger peninsula (Holtedahl, 1918, 1932; Siedlecka and Siedlecki, 1971; Harland and Gayer, 1972; Siedlecki, 1975).

Returning to the issue of the divergent Late Precambrian palaeopole, the difference in pole position between earlier published Late Precambrian data from northern Europe (cf. Fig. 3) and that of the present study can in fact be accounted for by invoking an anti-clockwise rotation of the northern part of the Varanger peninsula subsequent to the intrusion of the Båtsfjord dykes. This can quite easily be accomplished by a dextral horizontal movement along the Trollfjord–Komagelv Fault, assuming this fracture zone to have had a NW-NNW continuation. Such a trend has in fact been suggested by gravimetric investigations (Brooks, 1970). The NNW–SSE-trending fault system of Spitsbergen, and in particular the Billefjorden fault zone, may be highly relevant in this connection, but in the present study we have assumed that the NNW continuation of the Trollfjord–Komagelv–Ribachiy Fault originally passed somewhere to the west of Spitsbergen, possibly even up to and alongside the Sæfaxi Ridge in NE Greenland which bordered the comparatively deep "marginal trough" of Haller (1971). In this model the Billefjorden Fault zone is seen as a subparallel major fracture, the original extension of which may now be hidden beneath the Barents Shelf though not necessarily equivalent to the offshore fault depicted by Åm (1975).

On the assumption that the Trollfjord–Komagelv Fault is an ancient transform fracture, with a north-northwestward and southeastward continuation as sug-

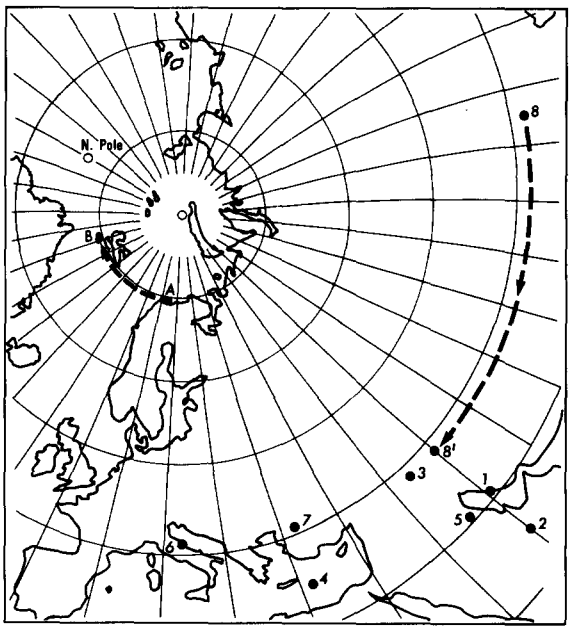


Fig. 4. Palaeomagnetic reconstruction of the Barents Sea Region of the Varanger peninsula. Poles *1–8* are the same as those of Fig. 3 but are here represented by their northern hemisphere equivalents. By applying a 65° clockwise rotation about a pole at 77°N, 60°E, the Varanger Peninsula Barents Sea Group (*A*) moves into a position (*B*) close to NW Spitsbergen. This movement shifts the Båtsfjord pole (*8*) to a new position (*8'*) which brings it into better agreement with the Late Precambrian polar group for NW Europe. Grid system is drawn with respect to the adopted pole of rotation (77°N, 60°E).

gested above, then the associated Euler pole will be situated just north of Novaya Zemlya, at around 77°N, 60°E (the alternative of linking the Trollfjord–Komagelv Fault with the Billefjorden lineament would only lead to a minor adjustment of this pole of rotation). As seen from Fig. 4, a 65° clockwise rotation

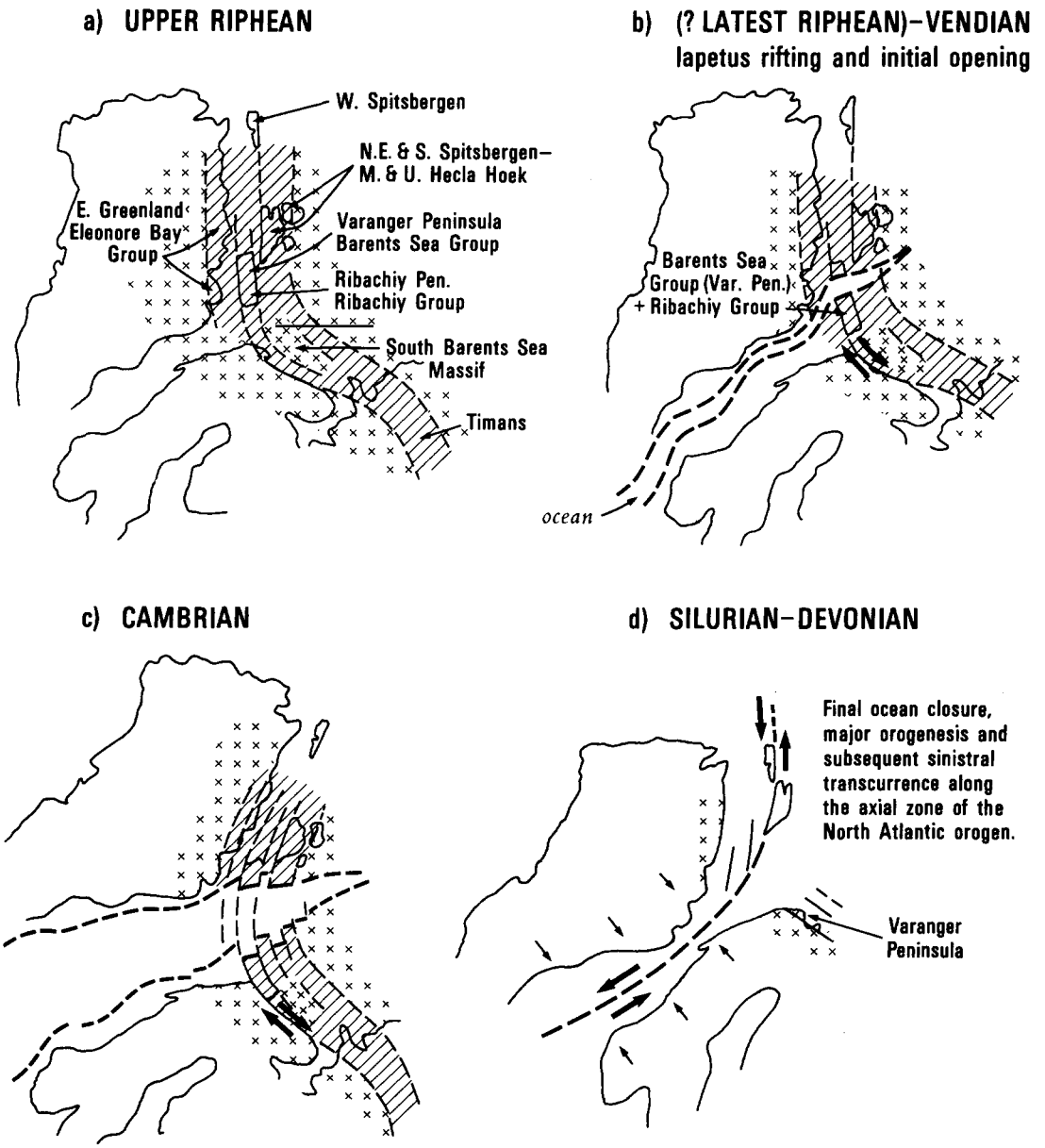


Fig. 5. Suggested palaeogeographical and megatectonic evolution at the junction between the Baltic, Barents and Laurentia cratons for the time interval Late Riphean–Late Devonian. The Late Riphean sedimentation (a) in fault-controlled basins (shaded) was followed by (b and c) rifting, Iapetus Ocean formation and "transcurrence" of the Varanger–Ribachiy block. After final ocean closure and orogenesis (d) a sinistral meagashear along the Billefjorden lineament in Late Devonian time juxtaposed the dissimilar Late Precambrian sedimentary sequences of west and east Spitsbergen, and is believed to have continued southwestwards along the axis of the Caledonian orogenic belt.

about this pole will bring the palaeopole for the Båtsfjord dykes into very good agreement with the Late Precambrian polar group for western Europe. Following such a rotational movement the geological evidence for a dextral transcurrent movement along the Trollfjord–Komagelv Fault is satisfied and the Varanger Peninsula Barents Sea Group will be located in the vicinity of the present-day position of NW Spitsbergen; i.e., in a location which is in line with the idea that the Barents Sea Group, the Eleonore Bay Formation (east Greenland) and the Hecla Hoek succession (east Spitsbergen) originated in a common depositional basin in the Upper Riphean. We suggest that also the Ribachiy Group sequence (Siedlecka and Siedlecki, 1967; Siedlecka, 1975) derives from the same depositional basin regime. Preliminary observations on sediment transport in the Barents Sea Group succession denote that currents were directed mostly towards the NE quadrant (D. Roberts, unpublished data, 1970; Siedlecka, 1972, in prep.), so that prior to the dextral transcurrence the maximum palaeoslope was probably inclined towards an easterly point indicating that the sequence was deposited in the western part of a miogeosynclinal trough.

The palaeomagnetic reconstruction outlined above, which is based on the present geographic position of Greenland/Spitsbergen versus the Baltic plate is equivalent to a transcurrent displacement of the order of 1000 km. However, this figure may well be an overestimate of the *actual* amount of displacement. For example, if Greenland/Spitsbergen and Fennoscandia were closer together at the time of transcurrent movement than they are at present, then the rotation required for establishing agreement between the Båtsfjord pole and palaeopoles of similar age in NW Europe can be accomplished by a smaller horizontal displacement. Because of the problems in re-establishing the ancient palaeogeography the displacement figure must in any case be given within fairly wide limits; between say 500 and 1000 km.

As shown in Fig. 5 we suggest that the separation of the Barents Sea/Ribachiy Groups from the Hecla Hoek and Eleonore Bay sequences is directly related to the formation of the Iapetus Ocean and reactivation of ancient "transverse" fractures. The suggested reconstruction for the Upper Riphean (cf. Fig. 5a) would also be in line with inferred palaeogeography based on palaeomagnetic comparison between Laurentia and the Baltic Shield (Ueno et al., 1975). The Varanger and Ribachiy successions were carried east-southeastwards, east of Iapetus, on the northern margin of the Baltic plate and dextrally displaced in relation to this craton (Fig. 5b, c). At the same time the east Spitsbergen and east Greenland sequences were drifting northwestwards, on the opposite side of the Iapetus Ocean, between the Laurentian and Barents cratons. Subsequent to the closure of Iapetus in the Arctic regions (Fig. 5d), and as a result of a changing compressional regime, a major sinistral transcurrence (between 200 and 1000 km) along the Billefjorden lineament juxtaposed the dissimilar Riphean–Lower Palaeozoic sequences of west and east Spitsbergen (Harland, 1972; Harland et al., 1974). This latter movement, which is dated to Svalbardian time (earliest late Devonian), appears to have propagated south- and southwestwards following the grain of the main Scandinavian–Greenland Caledonian orogenic belt, the fracture perhaps linking with the prominent Great Glen Fault of north Scotland which, in one school of thought, is also considered to provide evidence of large-scale sinistral displacement (Winchester, 1973; Storetvedt, 1974, 1975).

4. Discussion

Acceptance of the geophysical, geochronological and tectonic evidence implies that Siedlecka's (1975) model of in-situ or autochthonous Riphean to Vendian sedimentation in pericratonic and miogeosynclinal zones adjacent to the Fennoscandian or Baltic craton must be rejected or at least revised considerably (cf. Siedlecka and Siedlecki, 1977). Aspects of the model still hold, however, including the lateral strike-parallel macro-facies zonation from Varanger peninsula to the Timans, and the suggestion that the Trollfjord–Komagelv–Ribachiy Fault represents a fundamental ancient fracture (Siedlecka, 1975). The recognition of major, deep-seated, Precambrian fault zones in this part of Europe, especially along NW–SE trends, has indeed long been known; data from Scandinavia have been reviewed by Strömberg (1976). In the Timans and southern Barents Sea shelf region, geophysics has also indicated the existence of NW–SE-trending fracture zones which mark the SW and NE margins of the Timanian Baikalides and reach

down to the upper mantle, as well as delimiting a pre-Karelian South Barents Sea Massif (Fig. 5a) (Ersov et al., 1974; Maljavkin, 1974; Siedlecka, 1975). Raznitsyn (1972) has also recognized the old age of these fractures in referring to the sedimentary and orogenic belt of the Timans as representing an epikinegen, i.e. an elongate trough sited along pre-existing major basement fractures.

Harland (1972) and Harland et al. (1974) have suggested that a pre-existing N–S fracture system in the Svalbard region may have controlled the pattern of sedimentation in Riphean time with the Billefjorden Fault already present as a major fracture bordering the main basin. Thus, the geosynclinal trough which gave birth to the Riphean–lower Palaeozoic successions of east Greenland, east Spitsbergen and the Varanger Peninsula Barents Sea Group – the Ny Friesland Trough of Harland and Gayer (1972), or the Eleonore–Hecla Trough of Roberts and Gale (1977) – may have been of epikinegenic character, delimited by ancient, deep-seated fractures.

An important question is the age of the principal phase of dextral strike-slip translation along the Trollfjord–Komagelv Fault. Beckinsale et al. (1975) suggested a combination of:

(a) Post-640 m.y., pre-Early Ordovician.

(b) Post-Early Ordovician, pre-Early Silurian.

(c) Devonian, but pre-355 m.y. lateral movement.

Recent mapping on Nordkyn peninsula (D. Roberts, A. Siedlecka and E. Fareth, manuscript maps 1974–1975) and in NW Varanger peninsula by one of us (D.R.) has, on the other hand, disclosed that lateral movements along the Trollfjord–Komagelv Fault zone following the emplacement of the Kalak Nappe Complex (cf. Fig. 1) did not produce more than ca. 4–5 km and possibly only 1–2 km of displacement. As the principal Caledonian thermotectonic event in the Kalak Nappe Complex dates to 530–500 m.y. B.P., with thrusting largely of Late Cambrian–Early Ordovician but also of Silurian age (Sturt et al., 1975), we can narrow down the time of major transcurrent movement to post-640 and pre-500 m.y. B.P. However, how much of the right-lateral displacement was achieved during the actual Iapetus opening stage and how much in conjunction with the Late Cambrian–Early Ordovician Finnmarkian orogenic episode is virtually impossible to determine. With the Iapetus Ocean opening believed to date back to Vendian time (Roberts and Gale, 1977) it is tempting to relate the Båtsfjord dyke swarm in some way to the initial rifting and separation phase, perhaps in a "transtensional" regime (cf. Harland, 1971). As plate separation and ocean accretion proceeded along a general NE–SW trend the major NW–SE to NNW–SSE fault zones, which probably had been active intermittently through late Riphean time (including the Trollfjord–Komagelv–Ribachiy and Billefjorden lineaments), were probably reactivated and thus partly acted as loci for transform faults as spreading continued in the expanding oceanic area (Fig. 5). A similar development has been described by Wilson (1965) and other examples documented by Cebull et al. (1976) and Garson and Krs (1976). While the Trollfjord–Komagelv fracture zone can hardly be called a true transform fault, for part of its life it could be ranked as a "boundary transform" (Gilliland and Meyer, 1976) in as much as it is considered to have constituted a plate or subplate boundary.

The implications of the geotectonic reconstructions outlined above also have an important bearing on the regional context of the age of folding and metamorphism in the Barents Sea Group. On the basis of their K/Ar determinations Beckinsale et al. (1975) postulated the existence of a pre-640 m.y. B.P. deformation episode in this particular region, maintaining that the 500 m.y. B.P. Finnmarkian folding had not affected the Båtsfjord area. Radiometric work on phyllites from the Ribachiy Group of Ribachiy peninsula denotes a K/Ar whole-rock metamorphic age of somewhere between 700 and 900 m.y. B.P. (Konopleva, 1974). From our reconstructions this would denote that evidence of a comparable metamorphic event should be detectable in the east Greenland and Spitsbergen Riphean successions. Suggested, although as yet unfounded indications of an Upper Riphean metamorphism, at 700–800 m.y. B.P., have already been reported from Spitsbergen (Gayer et al., 1966); and Birkenmajer (1975) has described two important orogenic events from within the lower part of the Hecla Hoek succession, the Werenskioldian and Torellian diastrophisms. It is possible that the Werenskioldian may equate with the Carolinidian event of NE Greenland (Haller, 1971), whereas the Torellian could provide a link with the deformations recognizable on Varanger and Ribachiy and thus be representative of a Baikalian orogenic

event stretching from the Timans to the pre-Iapetus location of NE Greenland and Spitsbergen.

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